

Isaac Mubiru

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PROFESSIONAL SUMMARY

Backend engineer specialized in high-throughput event pipelines and zero-downtime migrations on Node.js + TypeScript. Built a [Pub/Sub event pipeline](#) processing millions of events at MTailor, cut \$5K/month in AWS spend by re-architecting the stack, and shipped an OpenAI + Vision OCR pipeline over WhatsApp Business API. Now applying the same throughput + observability discipline to LLM observability infrastructure at billion-trace scale.

CORE COMPETENCIES

TypeScript / Node.js (5+ yrs)

Event ingestion at scale

PostgreSQL & OLAP patterns

ClickHouse (ramping)

Pub/Sub & Kafka

Idempotent consumers

Cost optimization

OpenTelemetry & observability

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer (US Remote, reports to CTO)

- Designed and operated a **high-throughput Pub/Sub ingestion pipeline** for real-time MongoDB ↔ Firestore sync across 20+ apps — idempotent consumers, dead-letter queue, replay, zero data loss over 6 months of live production traffic.
- Re-architected production infra from AWS to Firebase/GCP — saved **\$5,000/month**, ran cost-modeled cutover with zero downtime.
- Built customer-facing 3D body visualization (ffmpeg + video overlay), measurably increasing buyer conversion.
- Migrated file storage from AWS S3 to GCS via a streaming Python pipeline with integrity checks across 20+ apps.
- Saved the company **\$5,000/month** by re-architecting services off AWS onto Firebase/GCP — designed cutover plan, ran shadow reads, executed without downtime.
- Implemented Express Shipping revenue feature end-to-end — backend pricing pipeline, UI, analytics — generating **+\$40 per order**.

Stack: TypeScript, Node.js, Firebase, GCP Pub/Sub, AWS, Python, ffmpeg, React

MTailor (Contractor)

Mar 2022 – Jul 2022

Backend Migration Contractor (US Remote)

- Hired via Upwork to execute backend migration from Parse/MongoDB to Firebase/GCP.
- Built migration script reading MongoDB and writing to Firestore with complex transformation logic.
- Documented new Firebase SDKs (Node + Python) and trained Ops team on the new dashboard + backup procedures.

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer (Singapore Remote)

- Built batch query layer reading **2M+ Firestore records** with paginated cursors + pre-aggregated views — kept Morningstar prices fresh in the PWA at low cost.
- Extended backend APIs using Firebase Cloud Functions + Express.js on Heroku.
- Shipped responsive UI pages with HTML, Tailwind, React.

Mind2Matter

Jan 2022 – Jul 2022

Software Engineer (US Remote, reports to CTO)

- Built backends for DeFi applications on Node.js + Web3.
- Delivered React UIs from Figma under tight timelines.

CodeBits

Jan 2020 – Jul 2021

Software Engineer / Tech Lead (Uganda Remote)

- Led a team of 4 building legal-tech systems for NGOs (FIDA Uganda, LASPNET Uganda).
- Architected a **microservices backend** with **Kafka + Docker + Kubernetes** for FIDA's case management app — service mesh with gRPC + Protobufs.
- Built USSD service for legal aid bridging to Node.js backend via gRPC.
- Built cross-platform mobile app (React Native + Expo, iOS + Android) with Firebase push.

PROJECTS

Parse → Firebase Migration (MTailor) zero downtime

Migrated 20+ production apps from Parse/MongoDB to Firebase/Firestore with dual-write Pub/Sub pipeline, per-app flag flip, and rollback discipline. Saved \$5K/month in AWS spend with no customer-visible incident.

Node.js, Pub/Sub, Firestore, MongoDB, GCP, AWS, Python

Healthcare WhatsApp Chatbot OpenAI + Vision

Built prescription-OCR chatbot on WhatsApp Business API integrating **OpenAI** + **Google Vision OCR**. Low-latency request flow with retry + observability around third-party LLM calls.

Node.js, OpenAI, Google Vision, WhatsApp Business API, GCP

Real-time Pub/Sub Sync Pipeline millions of events

Designed idempotent Pub/Sub consumer with dead-letter queue + replay for MongoDB ↔ Firestore sync. Pipeline ran parallel production traffic during 6-month migration with zero data loss.

Node.js, GCP Pub/Sub, Firestore, MongoDB, idempotency keys

Marketplace Recommendation Engine

Built collaborative + content-based filtering engine in MongoDB for service-provider matching. Ranked candidates from interaction history + skill graphs.

Node.js, MongoDB, vector similarity

EDUCATION

B.Sc. Computer Engineering — Makerere University

Aug 2018 – Sep 2022

Kampala, Uganda

CERTIFICATIONS

Open-source contributor & GitHub portfolio — github.com/zac-09

Ongoing

SKILLS

Languages: TypeScript, Node.js, JavaScript, Python, Go (intermediate), SQL

Backend / Infra: PostgreSQL, MongoDB, ClickHouse (ramping), GCP Pub/Sub, Kafka, Redis, S3, Docker, Kubernetes, AWS

Observability: OpenTelemetry (reading), custom freshness metrics, idempotent dead-letter pipelines

AI / Domain: OpenAI APIs, Google Vision, LLM integration patterns